

FL Simulation Results

File: test_A2

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 65.0% (0.65)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 65.0% (0.65) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8718
Clean + DP Protection Accuracy: 0.8719
Poisoned Accuracy: 0.8133
Poisoned + DP Protection Accuracy: 0.8140
Drop (Clean - Poisoned): 0.0584
Drop (Clean - Init): 0.0790
Drop (Clean DP - Init): 0.0791
Drop (Poisoned - Init): 0.0205
Drop (Poisoned DP - Init): 0.0212
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8778
Clean Recall: 0.8718
Clean F1 Score: 0.8711

Clean DP Precision: 0.8779
Clean DP Recall: 0.8719
Clean DP F1 Score: 0.8712

Poisoned Precision: 0.8229
Poisoned Recall: 0.8133
Poisoned F1 Score: 0.8125

DP Protection Precision: 0.8231
DP Protection Recall: 0.8140
DP Protection F1 Score: 0.8131

Summary

FL Simulation Complete
Task: 205d0c59-e048-49a6-859c-56e0edfa3b32
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8718
Clean DP Accuracy: 0.8719
Poisoned Accuracy: 0.8133
Data Poison Protection Accuracy: 0.8140
Drop (Clean - Poisoned): 0.0584
Drop (Clean - Init): 0.0790
Drop (Clean DP - Init): 0.0791
Drop (Poisoned - Init): 0.0205
Drop (Poisoned_DP - Init): 0.0212
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8778
Clean Recall: 0.8718
Clean F1 Score: 0.8711
Clean DP Precision: 0.8779
Clean DP Recall: 0.8719
Clean DP F1 Score: 0.8712
Poisoned Precision: 0.8229
Poisoned Recall: 0.8133
Poisoned F1 Score: 0.8125
DP Protection Precision: 0.8231
DP Protection Recall: 0.8140
DP Protection F1 Score: 0.8131